README

Numerical Experiments with Baseline Model

1 Model

1.1 Objective

$$\min_{y,q,O,U} \quad \underbrace{\sum_c F_c y_c}_{\text{entry costs}} + \underbrace{\sum_c K_c q_c}_{\text{vehicle costs}} + \frac{1}{|S|} \sum_{s\in S} \sum_{c\in C} \Big[\underbrace{c_c^+ O_c^s}_{\text{unused capacity}} + \underbrace{c_c^- U_c^s}_{\text{foregone margin}} \Big].$$

The superscripts in c_c^+ and c_c^- denote *excess* and *shortage*, not the sign of the cost. Scenarios are weighted equally $(1/|S|)$.

1.2 Constraints

$$\begin{aligned}
&0 \le q_c \le \bar{q}_c\, y_c && \forall c \in C,\\
&\textstyle\sum_c q_c \le Q,\\
&\kappa_c q_c - M_c p_c^s = O_c^s - U_c^s && \forall c \in C,\ s \in S,\\
&y_c \in \{0,1\}, \quad q_c \in \mathbb{Z}_{\ge 0},\\
&O_c^s,\, U_c^s \ge 0.
\end{aligned}$$

Constraint	Meaning
$0 \le q_c \le \bar{q}_c y_c$	No vehicles in a city that is not entered; up to the city-specific limit if it is entered.
$\sum_c q_c \le Q$	Total fleet limit.
$\kappa_c q_c - M_c p_c^s = O_c^s - U_c^s$	Service capacity minus robotaxi demand equals unused capacity minus unmet demand.

2 Numerical experiments with synthetic data (summary)

Two synthetic instances were run to answer two questions: (1) is the implementation correct and does it produce meaningful results, and (2) how sensitive are the results to parameter choices. Full details are in `numerical_experiments.tex` (small instance) and `annual_experiments.tex` (annual instance).

2.1 Small case instance (3 cities [A,B,C], $Q = 4$, 10 hand-designed scenarios)

Inputs. $M = (80, 100, 120)$, $\kappa = 10$, $\bar{q} = 2$, $F = 20$, $K = 5$, $c^+ = 0$, $c^- = 10$; mean choice probability 0.38 in every city.

Correctness. The MILP (SciPy/HiGHS) and full enumeration of all 60 feasible (y, q) give the same unique optimum $q = (0, 2, 2)$, $Z = 858$. Eight edge cases behave as expected: no vehicles, no demand, $c^- = 0$, or a prohibitive F give no entry; $Q = 8$ stops at the city limits $(2, 2, 2)$; $F = 0$ enters all three cities; demand equal to one vehicle's capacity gives $(1, 1, 1)$ with $U = O = 0$.

Meaningful results. The model concentrates in the two larger markets: entering A would serve 0.8 more trips (margin 8) but costs $F = 20$, so Z rises by 12. Service rate 30%, utilization 85.5%, improvement over no entry $\Delta Z = 282$.

Sensitivity. Switching points are identified rather than only three levels: three cities for $F < 8$, B and C for $8 < F < 160$, C alone for $160 < F < 162$, no entry above; three cities from $\kappa \geq 12.5$ (tie at 12); $(2, 0, 2)$ from $M_A \geq 1.3\times$ (tie at 1.25); c^+ leaves the allocation unchanged up to 15 and switches to $(1, 1, 2)$ from 15.5; $Q \geq 6$ is capped by the city limits. Without demand variability three allocations tie; with it only $(0, 2, 2)$ is optimal, and mean-demand allocations cost 0–16 more under the scenarios.

2.2 Large case instance (3 cities [A,B,C], $Q = 1{,}000$, 10 random scenarios)

Inputs. Literature-based proxies: $M = (577.9, 770.5, 963.2)$ million trips/year, $\kappa = 8{,}760$, $\bar{q} = 500$, $F = 3.5$M, $K = 7{,}972$, $c^+ = 0$, $c^- = 10$; p_c^s drawn independently from $\{0.002, 0.005, 0.01, 0.02\}$ so that demand and capacity are of the same order.

Correctness. Every solve was cross-checked against an independent dynamic program that also counts tied optima (the program itself was checked against enumeration on the small instance); all edge cases pass.

Meaningful results. Optimum $q = (132, 368, 500)$ with 72 ties in the B–C split; $\Delta Z = 60.85$M, service rate 34.9%, utilization 90.6%. The allocation follows the value of one more vehicle, $87{,}600 \times P(\text{demand} > \text{capacity}) - 7{,}972$: A is filled only while all ten scenarios have unmet demand. Entering A is worth only 0.39M more than $(0, 500, 500)$. With the reference choice probabilities (≥ 0.02) demand exceeds capacity in every scenario and any two cities tie regardless of M and p; demand differences matter only near $p \approx \kappa\bar{q}/M$.

Sensitivity. A drops out when $F > 3.89$M, $\kappa \leq 8{,}212.5$, or demand grows by $\geq 6.25\%$; A enters only for $Q \geq 1{,}000$ and the city limits bind at 1,500; $\bar{q} \geq 750$ gives $(0, 440, 560)$; $c^- = 2, 5, 10$ give one, two, three cities. c^+ changes nothing up to 15 (it only lowers ΔZ by $827{,}397\, c^+$) and shrinks the fleet to 528 at 40; it is equivalent to raising K by $c^+\kappa$ and c^- by c^+. Mean-demand allocations cost 0.39–8.14M more under the scenarios, and 50-scenario samples give seed-dependent allocations while 500-scenario samples are stable.

2.3 Which Inputs Matter: Decision versus Performance Sensitivity

The large instance separates two kinds of sensitivity: inputs that change the optimal allocation, and inputs that change the improvement ΔZ. The two lists differ.

Table 1: Inputs That Change the Optimal Allocation (Baseline $(132, 368, 500)$)

Input	Range examined	Effect on the allocation
Q, $\bar{q}$ (fleet and city limits)	0–2,000 / 100–1,500	The limits dictate the allocation: C only for $Q \leq 500$, B added at 600, A added at 1,000, saturation at 1,500; $\bar{q} \geq 750$ gives $(0, 440, 560)$.
κ (common)	$\pm 25\%$	A is dropped once κ falls by about 3% (switch near 8,498).
Demand $M \cdot p$ (common)	$\times 0.75$–1.25	A is dropped at +4.2%; smaller demand moves vehicles into A (170 at $\times 0.75$).
c^- (common)	2–20	One city at 2, two cities at 5, three cities from 10.
α (variability)	0–1	Two cities up to 0.7, three cities from 0.8.
Differences in one city	$\pm 25\%$	κ_c, c_c^-, M_c, and F_A move the allocation substantially; even small differences ($c_c^+ = 1$, $K_c \pm 25\%$) break the B–C tie.

Table 2: Inputs That Leave the Optimal Allocation Unchanged

Input	Range examined	Why the allocation is unchanged
K (common)	$\pm 25\%$	Every vehicle's value falls by the same amount in all cities, so the ordering is unchanged; Q remains binding.
c^+ (common)	0–15 (switch near 35)	Same mechanism; c^+ can be absorbed into $K + c^+\kappa$ and $c^- + c^+$.
F (common)	up to +11%	Only above 3.89 million is A dropped; B and C are entered at every F examined.
M_A	0.875–1.25	Only A's fleet adjusts (116–165 vehicles); the structure is unchanged.

Table 3: Sensitivity of the Improvement ΔZ (USD Million), Most to Least Sensitive

Input	Range examined	ΔZ over the range
c^{-}	2–20	0.29 to 140.18
κ	$\pm 25\%$	44.52 to 77.25
Q, $\bar{q}$	0–1,500 / 100–1,500	0 to 84.12 / 13.39 to 61.95
F, K	$\pm 25\%$	linear: ± 2.4 (F), ± 2.0 (K)
$M \cdot p$ multiplier	$\times 0.75$–1.25	57.15 to 63.78
c^{+}	0–15	-0.83 per USD of c^{+}
M_A	$\times 0.75$–1.25	60.46 to 61.43 (almost flat)